

- [About](#)
- [Online book](#)
- [Python Exercises](#)
- [Video Lectures](#)
- [Teaching Material](#)

2.4 Summary

The Hodgkin-Huxley model describes the generation of action potentials on the level of ion channels and ion current flow. It is the starting point for detailed biophysical neuron models which in general include more than the three types of currents considered by Hodgkin and Huxley. Electrophysiologists have described an overwhelming richness of different ion channels. The set of ion channels is different from one neuron to the next. The precise channel configuration in each individual neuron determines a good deal of its overall electrical properties.

Literature

A nice review of the Hodgkin-Huxley model including some historical remarks can be found in Nelson and Rinzel [1995]. A comprehensive and readable introduction to the biophysics of single neurons is provided by the book of Christof Koch (269). Even more detailed information on ion channels and non-linear effects of the nervous membrane can be found in B. Hille's book on 'Ionic channels of excitable membranes' (221). The rapidly growing knowledge on the genetic description of ion channel families and associated phenotypes is condensed in Channelpedia (419).

Exercises

- 1. Nernst Equation** . Using the Nernst equation (Eq. 2.2) calculate the reversal potential of Ca^{2+} at room temperature (21 degree Celsius), given an intracellular concentration of 10^{-4} mM and an extracellular concentration of 1.5 mM.
- 2. Reversal Potential and Stationary Current-Voltage Relation** . An experimentalist studies an unknown ion channel by applying a constant voltage u while measuring the injected current I needed to balance the membrane current that passes through the ion channel.
 - a) Sketch the current-voltage relationship (I as a function of u) assuming that the current follows $I_{\text{ion}} = g_{\text{ion}} m h (u - E_{\text{rev}})$ with $g_{\text{ion}} = 1$ nS and $E_{\text{rev}} = 0$ mV where $m = 0.1$ and $h = 1.0$ are independent of the voltage.
 - b) Sketch qualitatively the current-voltage relationship assuming that the current follows $I_{\text{ion}} = g_{\text{ion}} m h (u - E_{\text{rev}})$ with $g_{\text{ion}} = 1$ nS and $E_{\text{rev}} = 0$ mV where $m_0(u)$ and $h_0(u)$ have the qualitative shape indicated in Fig. 2.16.
- 3. Activation Time Constant** . An experimentalist holds the channel from Fig. 2.16 A and B at $u = -50$ mV for two seconds and then suddenly switches to $u = 0$ mV. Sketch the current passing through the ion channel as a function of time assuming $I_{\text{ion}} = g_{\text{ion}} m h (u - E_{\text{rev}})$ with $g_{\text{ion}} = 1$ nS and $E_{\text{rev}} = 0$ mV.
- 4. The Power of the Exponent** . An experimentalist holds an unknown potassium ion channel with activation variable n with voltage dependence $n_0(u)$ and time constant τ_n at $u = -50$ mV for two seconds and then, at time $t = 0$, suddenly switches to $u = 0$ mV.
 - a) Sketch the activation variable n , n^2 , n^3 as a function of time for times smaller than τ_n .
 - b) Show mathematically that for $0 < t < \tau_n$ the time course of the activation variable can be approximated $n(t) = n_0(-50\text{mV}) + [n_0(0\text{mV}) - n_0(-50\text{mV})] t / \tau_m$
 - c) Do you agree with the statement that "the exponent p of in the current formula $I_{\text{ion}} = g_{\text{ion}} n^p (u - E_{\text{rev}})$ determines the 'delay' of activation"? Justify your answer.
- 5. Hodgkin-and-Huxley Parameter Estimation** . Design a set of experiments to constrain all the parameters of the two ion channels of the Hodgkin-Huxley model. Assume that the neuron has only the I_{Na} and I_{K} currents and that you can use tetrodotoxin (TTX) to block the sodium ion channel and tetraethylammonium (TEA) to block the potassium ion channel.

$\alpha_m(u) = 1 / [1 - e^{-(u+a)/b}]$ and $\beta_m(u) = 1 / [1 - e^{-(u+a)/b}]$ (compare Table 2.1), the stationary value of the activation variable can be written as $m_0(u) = 0.5 [1 + \tanh[\beta(u - \theta_{\text{act}})]$. Determine the activation threshold θ_{act} and the activation slope β .

Hint: $\tanh(x) = [\exp(x) - \exp(-x)] / [\exp(x) + \exp(-x)]$.